higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1

$$X_C = \frac{1}{2\pi \times 50 \times 45{,}5 \times 10^{-6}}$$ ✓

$\therefore X_C = 69.958\ \Omega$ ✓

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$ ✓

$$= \sqrt{10^2 + (16 - 70)^2}$$

$\therefore Z = 54{,}877\ \Omega$ ✓

$$I_T = \frac{V_{AC}}{Z}$$

$$= \frac{220}{54.877}$$ ✓

$\therefore I_T = 4{,}009\ A$ ✓ (6)

1.1.2 $V_R = IR = 4{,}009 \times 10$ ✓ $= 40{,}090\ V$ ✓
$V_L = IX_L = 4{,}009 \times 16$ ✓ $= 64{,}144\ V$ ✓
$V_C = IX_C = 4{,}009 \times 69.958$ ✓ $= 280{,}462\ V$ ✓ (6)

1.2 1.2.1 B
1.2.2 C
1.2.3 D
1.2.4 E
1.2.5 A

(5 × 1) (5)

1.3

$$F = \frac{R}{2\pi \times 20 \times 10^{-6}}$$ ✓

$$= \frac{2{,}5 \times 10^{3}}{2\pi \times 20 \times 10^{-6}}$$ ✓

$= 19{,}894\ kHZ$ ✓ (3)

[20]

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QUESTION 2

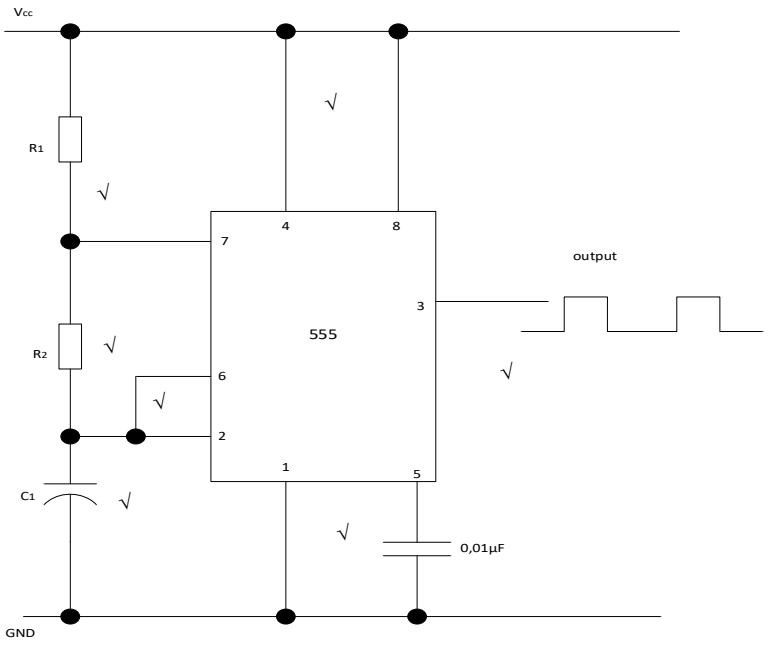
2.1 $f_o = \frac{1}{2\pi\sqrt{LC}}$

$f_o = \frac{1}{2\pi\sqrt{80^{-6} \times 900^{-12}}}$✓

f_o = 593,121 kHz✓ (2)

2.2
- Circuit components
- Supply voltage
- Output load
- Operating point of active device
- Stray capacitance (5)

2.3



(7)

2.4 2.4.1 UPS (uninterrupted power supply)
2.4.2 Transformer
2.4.3 Switching mode
2.4.4 Transient

(4 × 1) (4)

2.5 2.5.1 A device that supplies electrical power to a load. (1)
2.5.2 A DC battery source is used to develop a 220V AC output. (2)
2.5.3 DC needs to change to AC. (1)

2.6
- Break-over voltage√ – if the voltage across the SCR exceeds the rated forward break-over voltage, √ the SCR will start conducting due to avalanche breakdown. √
- Gate triggering√ – the SCR is operated with an anode voltage slightly less than the rated forward break-over voltage √and is triggered into conduction by low-power gate pulse. √ (2 × 3) (6)

2.7
- The output current flows during the entire cycle of the AC input signal.
- The operation of the amplifier is restricted to the smaller central region of the load line.
- Because the transistor operates over the linear region of the load line, the output waveform is similar to the input.
- It has an efficiency of between 25 and 50%. (Any 2 × 1) (2)

[30]

QUESTION 3

3.1
- The replacement transistor should have a V_{CBO} rating equal to or higher that of the original.
- The replacement transistor should have V_{CEO} rating equal to or higher that of the original.
- The replacement transistor should have V_{EBO} rating equal to or higher that of the original.
- The replacement transistor should able to handle as much or more collector current than the original.
- The replacement transistor should able to dissipate as much or more power as the original.
- The replacement transistor should have a gain equal to or better than the original.
- The replacement transistor should have a transition frequency equal to or faster than the original transistor and it should be as close to the original specification as possible. (7)

3.2
- The strip of copper track can loosen after components have been changed several times.✓
- When a copper track breaks, it may leave an invisible hairline of copper.✓ This results in an intermittent connection on the track, possibly causing a short circuit.✓
- The relationship between components and circuit diagram may be confusing for beginners✓ because the connecting tracks and where they are broken cannot be seen from the component side. This means that you have to turn the board over when doing fault-finding.✓ (5)

3.3
- Forward resistance
- Reverse resistance
- Reverse current (Any 2 × 1) (2)

3.4 3.4.1
- The power supply is OFF.
- The ammeter is faulty.
- There is a possible break between the resistors (open circuit).

3.4.2
- R1 is shorted out
- R2 is shorted out
- R3 is shorted out

(3 × 2) (6)
[20]

QUESTION 4

4.1
- Incorrect hardware installation or insufficient hardware
- Malware
- Hardware conflicts
- Freeware/Trialware
- Incomplete installation
- Uninstalled programs
- Software conflicts

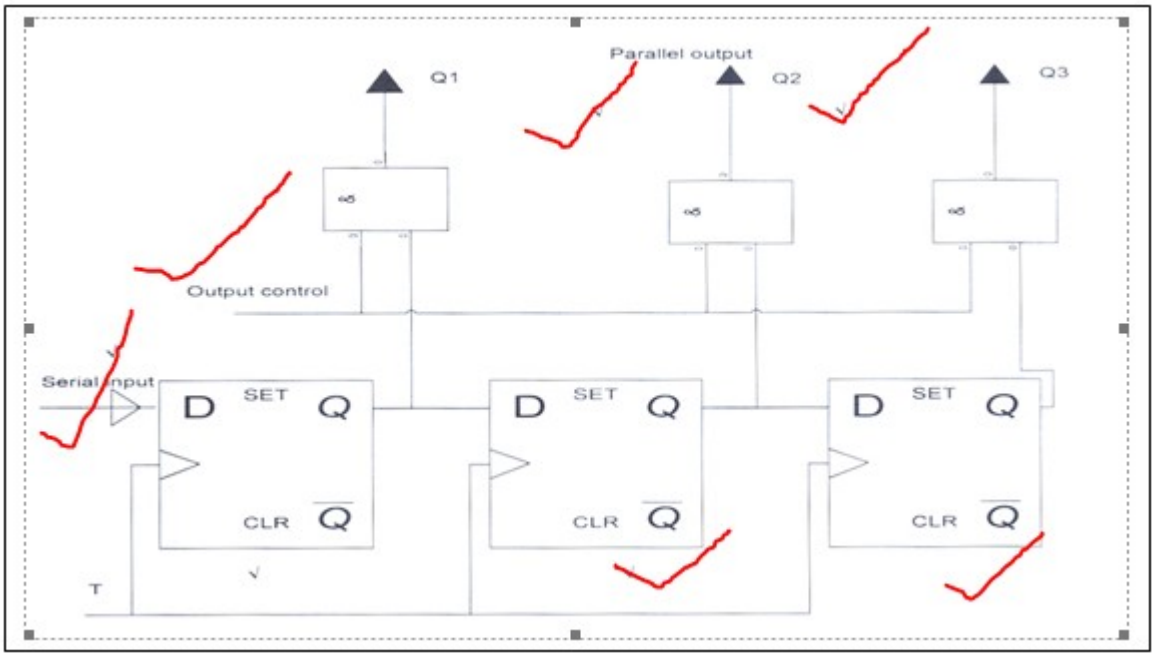
(Any 5 × 1) (5)

4.2 4.2.1 0,159230 exponent is a 6-mantissa
0,159230 × 10^6 – floating point number

4.2.2 Binary number 11001011 to floating point number.
1,1001011 × 2^6

(Any 2 × 2) (4)

4.3

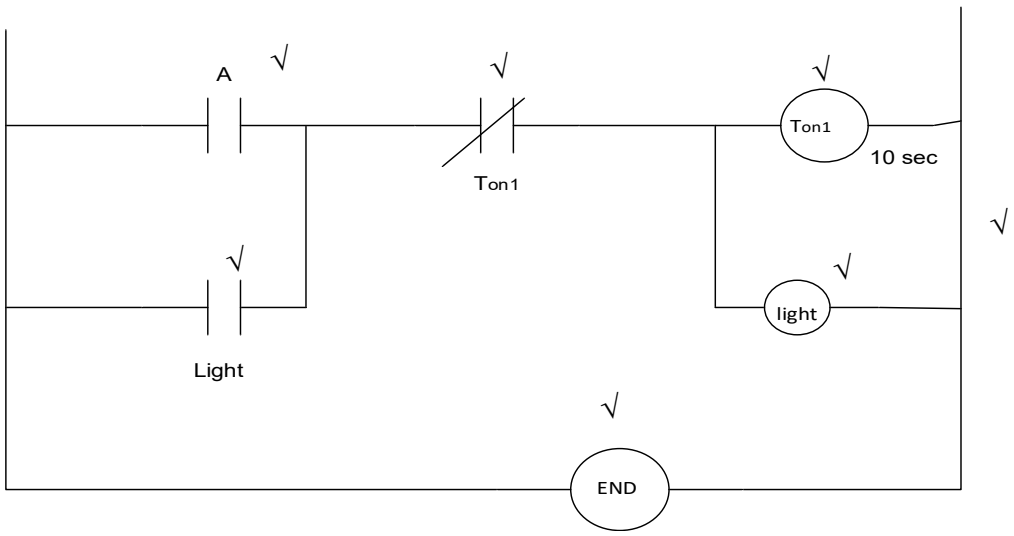


(6)
[15]

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QUESTION 5

5.1



(7)

5.2 5.2.1

- Mechanical and magnetic switches
- Photoelectric sensors
- Optical encoders
- Strain gauges
- Keypads
- Pressure sensors (Any 3 × 1) (3)

5.2.2

- Execute instructions
- Check the states of the outputs
- Calculate and store values in the variable data memory
- Fetch I/O information and data from the variable data memory
- Fetch instruction from the user program memory (Any 2 × 1) (2)

5.3 5.3.1 True
5.3.2 True
5.3.3 True

(3 × 1) (3)
[15]

TOTAL: 100

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